# BAYESIAN MODELING OF COMPLEXITY

## Bayesian Hierarchical Models II

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# Agenda


1 Recap & Today

2 Advanced Modeling Techniques

- Introduction
- Centered vs. Non-Centered Parameterization
- Crossed Random Effects

3 Practical Considerations

4 Wrap-up & Next

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# Recap

**What we covered last session:**

- Why hierarchical models? (Partial pooling)
- Basic structure: Data $\rightarrow$ Parameters $\rightarrow$ Hyperparameters
- Multi-level linear models
- Random intercepts vs. random slopes
- ICC (Intraclass Correlation)

**Key concept:** Shrinkage $\rightarrow$ small groups borrow strength from global pattern

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# Advanced Topics

**Advanced Modeling:**

- Non-centered parameterization
- Non-nested models (crossed random effects)
- Designing and calibrating hierarchical ABMs

**Diagnostics:**

- Divergences in hierarchical models
- Effective sample size
- Posterior predictive checks

**What we cover:**

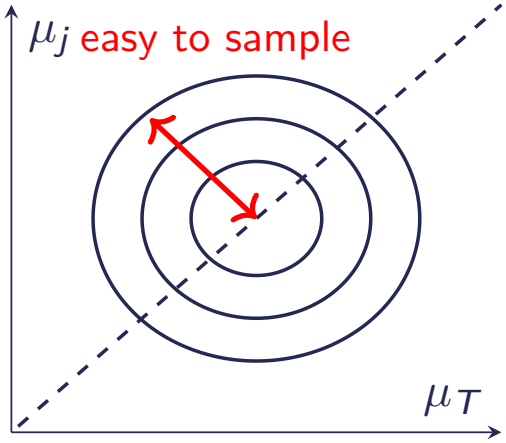
- Non-centered parameterization
- Non-nested models (crossed random effects)
- Diagnostics

# The Funnel Problem

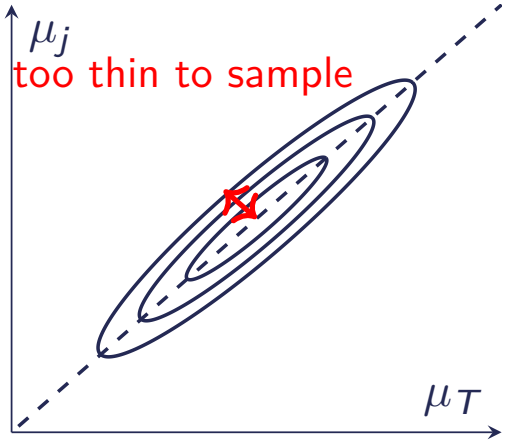
**Challenge in hierarchical models:** Geometry can cause MCMC problems

**Centered parameterization:**
$$y_j \sim \mathcal{N}(\mu_j, \sigma^2)$$
$$\mu_j \sim \mathcal{N}(\mu_T, \tau^2)$$

Large $\tau$




Small $\tau$




When $\tau \to 0$, group means satisfy $\mu_j \approx \mu_T$.

$\to$ **MCMC gets stuck!** Hard to move across narrow correlated regions.

# Non-Centered Parameterization

**Problem:** Divergence due to funneling

**Solution:** Reparameterize the model (?)

**Non-centered parameterization:**

$$y_j \sim \mathcal{N}(\mu + \tau \cdot \eta_j, \sigma^2)$$
$$\eta_j \sim \mathcal{N}(0, 1)$$

where $\alpha_j = \mu + \tau \cdot \eta_j$

**Key difference:**

- $\eta_j$ always has same scale (standard normal)
- $\tau$ acts as multiplier
- No geometry changes with $\tau \rightarrow$ easier sampling!

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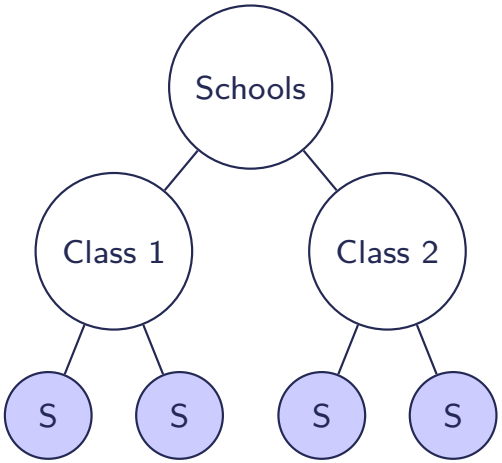
# Nested vs. Non-Nested Hierarchies

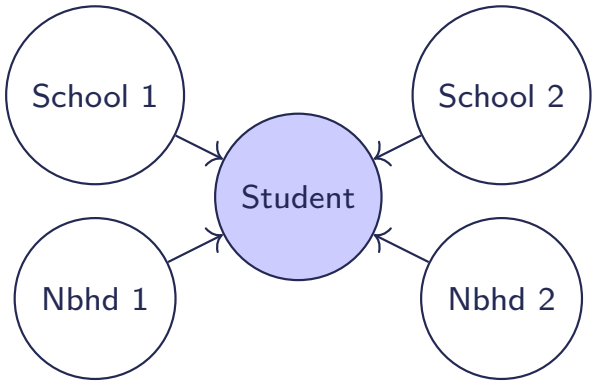
**Nested (tree structure):**

- Each lower level belongs to exactly one upper level
- Clear parent-child relationships
- students → classes → schools




**Non-Nested (crossed):**

- Lower levels can belong to multiple upper levels
- Network structure
- students → (schools & neighborhoods)




**Key difference:**

- Nested: One random effect per level
- Non-nested: Multiple random effects at the same level (can interact!)

# Beyond Nested: Crossed Effects

**Crossed random effects model:**

$$y_i = \mu + \alpha_{\text{school}[i]} + \beta_{\text{neighborhood}[i]} + \epsilon_i$$

where
$$\begin{array}{lll} \alpha_j & \sim \mathcal{N}(0, \sigma_\alpha^2) & j = 1, \ldots, J_{\text{schools}} \\ \beta_k & \sim \mathcal{N}(0, \sigma_\beta^2) & k = 1, \ldots, K_{\text{ngbs}} \end{array}$$

**Key feature:** Two independent random effects at same level

**Questions we can answer:**

- Which matters more: school or neighborhood? ($\sigma_\alpha$ vs. $\sigma_\beta$)
- What's the outcome for the student in school $j$ & ngb. $k$?
- Can add interaction: $\gamma_{jk}$ (but need lots of data!)

# Example: Student Achievement Crossed Model

**Scenario:** Students attend schools but live in neighborhoods

**Model:**

$$\text{TestScore}_{ijk} \sim \mathcal{N}(\mu_{ijk}, \sigma^2)$$
$$\mu_{ijk} = \mu + \alpha_j + \beta_k + X'_{ijk}\gamma$$
$$\alpha_j \sim \mathcal{N}(0, \sigma_\alpha^2) \quad \text{(school effect)}$$
$$\beta_k \sim \mathcal{N}(0, \sigma_\beta^2) \quad \text{(neighborhood effect)}$$

**Data structure:**

- Student $i$ in school $j$ and neighborhood $k$
- Not all (school, neighborhood) combinations exist

**Questions:**

- Variance decomposition: $\frac{\sigma_\alpha^2}{\sigma_\alpha^2+\sigma_\beta^2+\sigma^2}$ = school effect
- Predictions for new (school, neighborhood) pairs
- Which intervention point more effective?

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# Challenges in Hierarchical Models

**Common problems:**

1. Divergences (funnel geometry)
2. Low ESS
3. Non-convergence ($\hat{R}$)

**Checks:**

1. Group-level predictions
2. Individual-level predictions

**What to look for:**

- Generate and compare data for each group and for individuals within groups
- Check the posterior predictive distribution

**In PyMC:**

- `pm.sample_posterior_predictive()`
- `az.plot_ppc()` for visualization

# When Hierarchical Models Fail

**Signs of trouble:**

1. Divergences won't go away (even after reparameterization)
2. $\sigma_\alpha \to 0$ (groups are identical $\to$ use pooling!)
3. $\sigma_\alpha$ very large (groups completely different $\to$ no pooling!)
4. Very few groups ($J < 5$) $\to$ can't estimate hyperparams well

**Possible Solutions:**

- **Too few groups:** Fix hyperparameters (empirical Bayes) or use stronger priors
- **No variation/Too much variation:** Use complete pooling/ no pooling
- **Divergences:** Increase *target_accept*, non-centered parameterization
- **Model too complex:** Simplify

**Key principle:** Match model's complexity to data richness

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# Wrap-up

**Hierarchical Models:**

- Why hierarchical models (partial pooling)
- Basic structure and notation
- Random intercepts and slopes
- Nested vs. non-nested (crossed)
- Non-centered parameterization
- Diagnostics

**Complete workflow for your research:**

1. **Identify structure:** What are the groups?
2. **Calculate ICC:** Is grouping important?
3. **Start simple:** Random intercept model
4. **Check diagnostics:** $\hat{R}$, ESS, divergences, PPC
5. **Add complexity:** Random slopes/intercepts etc, if needed
6. **Compare models:** LOO-CV
7. **Validate:** Posterior predictive checks
8. **Interpret:** Variance components, shrinkage, predictions

# What's Next?

**Course continuation:**

- Your paper
- Your ABM project presentations

**You can now:**

- Design ABMs and embed Bayesian reasoning in it
- Calibrate ABMs to data
- Model data using Bayesian simple and hierarchical models

**Continue learning:**

- Practice with your project
- Explore PyMC documentation

# Questions?

Thank you for your engagement throughout the semester!